Set Up Your Own Network Storage Using FreeNAS

This brief overview of FreeNAS, an advanced FreeBSD-based operating system for network-attached storage servers, is followed by a few simple steps to help you install it in a virtual machine.

A NAS is usually a dedicated hardware device that connects to a LAN, where file-level computer data can be stored and retrieved across that network. It uses file-sharing network protocols such as NFS and CIFS (Common Internet File System) or SMB (Server Message Block) to manage file operations. It is often without peripherals (no keyboard and display) and supports hard drives, virtual drives, or tape drives. It can be connected, controlled and configured over the network, using popular browsers or consoles. NAS devices work perfectly well with stripped-down operating systems designed specifically for commodity PC hardware.

Network Attached Storage vs Storage Area Network

NAS is different from SAN (Storage Area Network). See Table 1 on the next page that lists some of the differences.

Use cases

- To provide centralised storage for client computers.
- To enable easy, simple and cost-effective load balancing and fault tolerance.
- Is most effective when a large amount of data needs to be managed.